

Http protocol

Hyper Text Transfer Protocol is an application protocol used to communicate and transfer data between a web browser(client) and a web server.

If you open <http://www.wikipedia.org>, your browser sends a request to the Wikipedia server, and the server sends the Wikipedia web page back to your browser.

Network Analysis using Wireshark – HTTP Protocol

In this practical, we will use Wireshark to capture and analyze HTTP packets exchanged between a web browser and a web server. This helps us understand how HTTP communication takes place on a network.

1. Make sure your computer is connected to the Internet through a Wi-Fi or Ethernet connection.
2. Open your web browser. It is recommended to clear the browser cache and cookies so that fresh HTTP packets are captured instead of cached data.
3. Open the Wireshark application.
4. Select the active network interface (Wi-Fi or Ethernet) that is currently connected to the Internet.
5. Click **Start** to begin packet capture. Wireshark will now capture all packets sent and received by your computer.
6. Open your web browser and enter the URL:
<http://gtu.ac.in>
(or any other HTTP website if the above site is unavailable).
7. Wait until the web page loads completely. During this process, your browser sends an **HTTP Request** to the web server, and the server sends an **HTTP Response** back to your browser. These packets are captured by Wireshark.
8. Return to Wireshark and click **Stop** to stop capturing packets. The captured packets will now be displayed in the packet list.
9. In the **Display Filter** bar, type:
10. `http`

and press **Enter** or click **Apply**. Only HTTP packets will now be displayed.

11. Select the first **HTTP GET** packet. In the packet details pane, observe the information such as:

- Source IP Address
- Destination IP Address
- HTTP Method (GET)
- Host
- Request URI

11. You can also filter only your computer's HTTP traffic by using the following filter:

```
http && (ip.src == <your_IP> || ip.dst == <your_IP>)
```

Replace <your_IP> with your computer's IP address. This filter displays only the HTTP packets exchanged between your computer and the web server.

For example, if your IP address is 169.1.19.87, then your filter should be `http && (ip.src == 169.1.19.87 || ip.dst == 169.1.19.87)`.

12. Observe the color coding used by Wireshark:

- **Green** – TCP packets
- **Dark Blue** – DNS packets
- **Light Blue** – UDP packets
- **Black** – TCP packets with errors or retransmissions

Result:

The HTTP Request and HTTP Response packets exchanged between the client (browser) and the web server are successfully captured and analyzed using Wireshark.

Experiment – 1

1. List up to IO different protocols that appear in the Protocol column in the unfiltered packet-listing window.
2. What is the Internet address of the www.gujaratuniversity.ac.in? What is the Internet address of your computer?

->nslookup www.gujaratuniversity.ac.in

->ipconfig

```
C:\WINDOWS\system32\ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    IPv6 Address. . . . . : 2402:3a88:4632:ef14:7581:ea42:77fc:1b6f
    Temporary IPv6 Address. . . . . : 2402:3a88:4632:ef14:75ad:6d8c:ee05:4f2f
    Link-local IPv6 Address . . . . . : fe80::4e01:15c3:b357:ac0c%7
    IPv4 Address. . . . . : 10.136.38.140
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::7430:22ff:fe05:f64%7
                                10.136.38.202

Wireless LAN adapter Local Area Connection* 2:

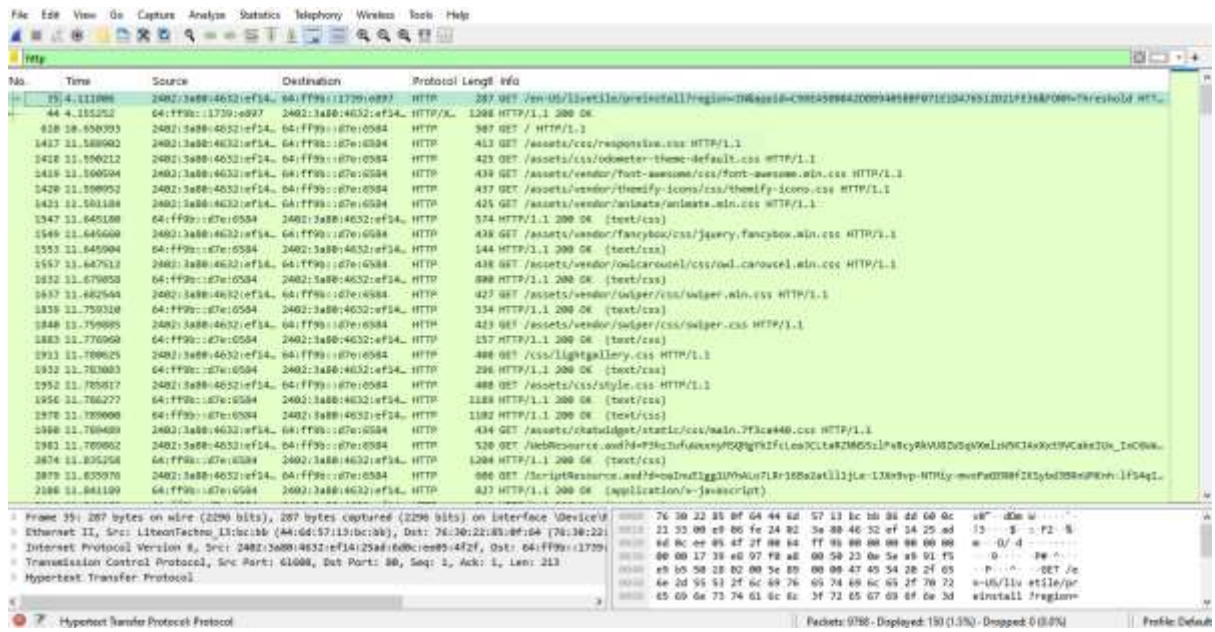
    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . . : fe80::c4d3:1248:9169:73e4%11
    IPv4 Address. . . . . : 192.168.137.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

```
C:\WINDOWS\system32>nslookup www.gujaratuniversity.ac.in
Server: Unknown
Address: 10.136.38.202

Non-authoritative answer:
Name:   gujaratuniversity.ac.in
Address: 64:ff9b::e8b:7a12
        14.139.122.18
Aliases: www.gujaratuniversity.ac.in
```

3. http protocol only with Wireshark running on your computer.

Ans: Total http:150



4. How many packets did you capture (total of all protocols, not just HTTP)?

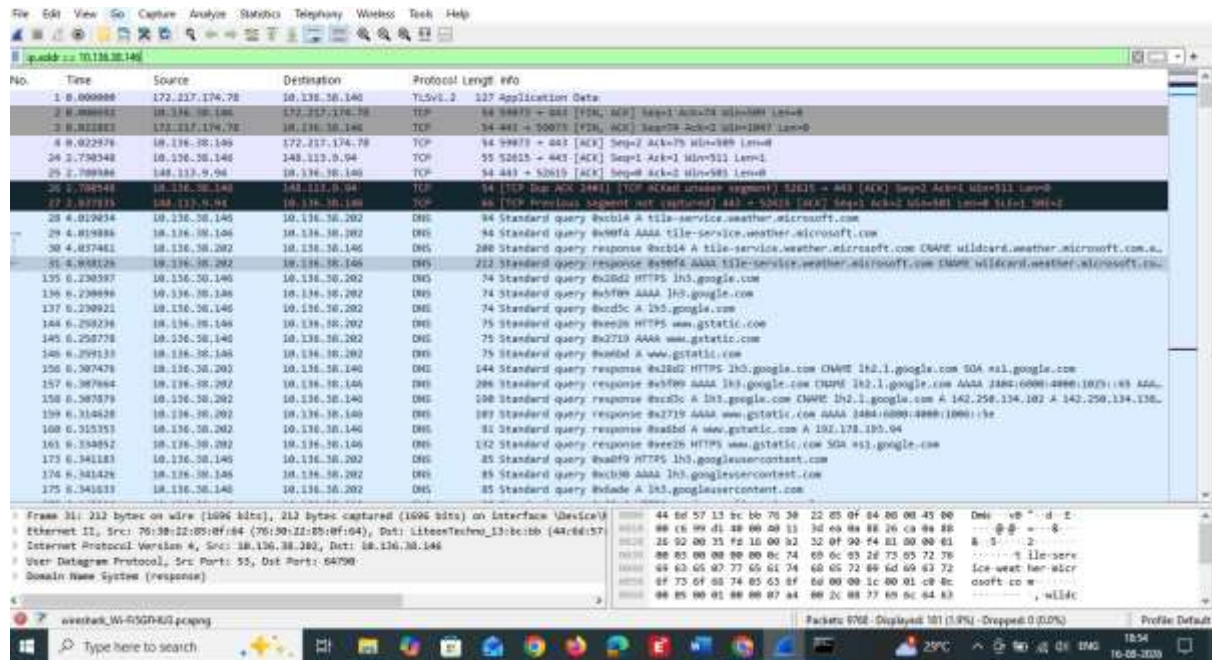
Ans :Total packets: 9768

[illegible]

5. Now, use display filters to determine how many packets contain your ip address (hint: Use ip.addr instead of the clumsy ip.src or ip.dst format).

Ans: ip.addr==10.136.38.146

Ip address :181 packets



6. What is this filter you used? Now, reverse the filter to determine how many packets don't contain your ip address.

Ans: !ip.addr==10.136.38.146

Ip address: 9587 packets

The image shows a Wireshark packet capture interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet analysis. The main window is divided into three panes:

- Packet List Pane:** Displays a list of captured packets. The first packet is selected, showing its details in the right pane.
- Packet Details Pane:** Shows the hierarchical structure of the selected packet, including Ethernet II, Internet Protocol Version 6, and Transmission Control Protocol.
- Packet Bytes Pane:** Displays the raw data of the selected packet in hexadecimal and ASCII.

The selected packet (No. 23) is a TCP segment from 2000::4700:4400::ac4 to 2402::3a00:4032:ef14. The details pane shows the following information:

- Ethernet II:** Src: 76:98:22:85:0f:64 (76:98:22:85:0f:64), Dst: LiteonTechno_13:bc:98 (64:00:57:13:bc:98)
- Internet Protocol Version 6:** Src: 2000::4700:4400::ac4, Dst: 2402::3a00:4032:ef14, Len: 60
- Transmission Control Protocol:** Src Port: 443, Dst Port: 51518, Seq: 1597, Ack: 3342, Len: 0

The packet bytes pane shows the raw data in hexadecimal and ASCII, with a filter applied to the data field.

By inspecting and looking at the information in the HTTP GET and response messages, answer the following questions.

7. What is the IP address of your computer? Of the gtu.ac.in server?
(nslookup gtu.ac.in)

```
C:\WINDOWS\system32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    IPv6 Address. . . . . : 2402:3a80:4632:ef14:7581:ea42:77fc:1b6f
    Temporary IPv6 Address. . . . . : 2402:3a80:4632:ef14:25ad:6d8c:ee05:4f2f
    Link-local IPv6 Address . . . . . : fe80::4e01:15c3:b357:ac8c%7
    IPv4 Address. . . . . : 10.136.38.146
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::7438:22ff:fe85:f64%7
                                10.136.38.202

Wireless LAN adapter Local Area Connection* 2:

    Connection-specific DNS Suffix . : 
    Link-local IPv6 Address . . . . . : fe80::c443:1248:0169:73e4%21
    IPv4 Address. . . . . : 192.168.137.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

C:\WINDOWS\system32>nslookup www.gtu.ac.in
Server: Unknown
Address: 10.136.38.202

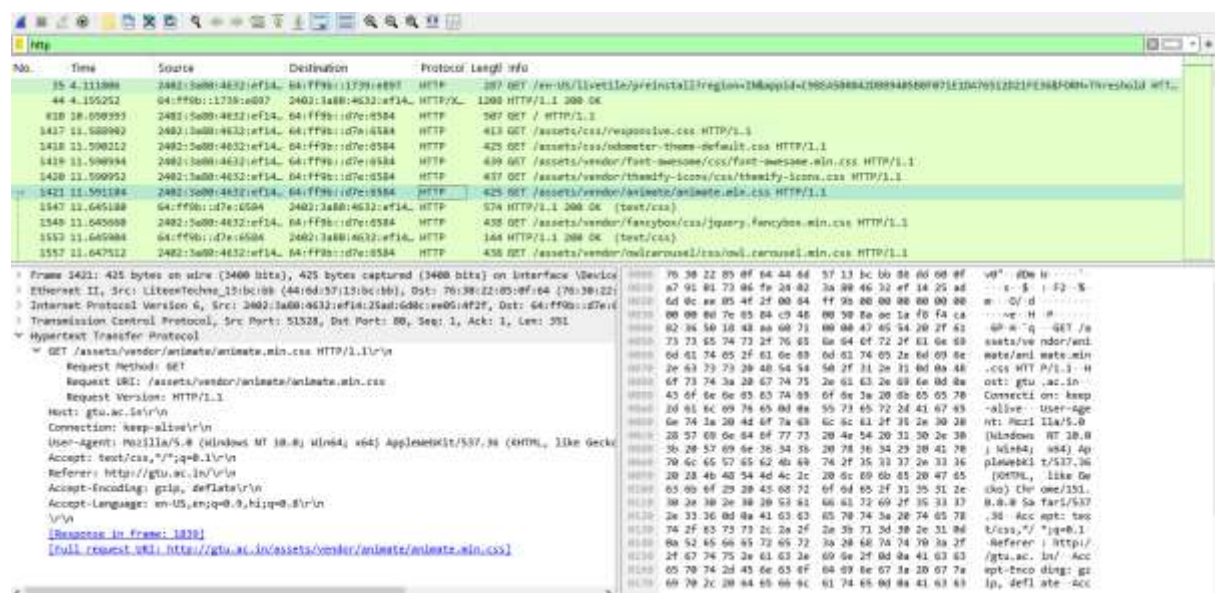
Non-authoritative answer:
Name:   www.gtu.ac.in
Addresses:  64:ff9b::deb:276d
            64:ff9b::d7e:6584
            13.126.101.132
            13.235.39.109
```

8. What HTTP version is your browser running? What version of HTTP is the server running?

(HTTP browser **request** line (GET / HTTP/1.1)& HTTP server **response** line (HTTP/1.1 200 OK)

(http.request.method == "GET") (http.response)

ANS : Request version: 1.1



Response version: 1.1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

hmp

No.	Time	Source	Destination	Protocol	Length	Info
35	4.111886	2402:3a00:4632:ef14::64	ff9b::1739:e897	HTTP	287	GET /en-US/livetile/preinstall?region=IN&appid=C98EA588642D66940588F073E1D476512D21FE
44	4.155252	64:ff9b::1739:e897	2402:3a00:4632:ef14::64	HTTP/X	1208	HTTP/1.1 200 OK
510	10.658393	2402:3a00:4632:ef14::64	ff9b::d7e:6584	HTTP	507	GET / HTTP/1.1
1417	11.588982	2402:3a00:4632:ef14::64	ff9b::d7e:6584	HTTP	413	GET /assets/css/responsive.css HTTP/1.1
1418	11.590212	2402:3a00:4632:ef14::64	ff9b::d7e:6584	HTTP	425	GET /assets/css/odometer-theme-default.css HTTP/1.1
1419	11.590594	2402:3a00:4632:ef14::64	ff9b::d7e:6584	HTTP	459	GET /assets/vendor/font-awesome/css/font-awesome.min.css HTTP/1.1
1420	11.590912	2402:3a00:4632:ef14::64	ff9b::d7e:6584	HTTP	457	GET /assets/vendor/themify-icons/css/themify-icons.css HTTP/1.1
1421	11.591184	2402:3a00:4632:ef14::64	ff9b::d7e:6584	HTTP	425	GET /assets/vendor/animate/animate.min.css HTTP/1.1
1547	11.645100	64:ff9b::d7e:6584	2402:3a00:4632:ef14::64	HTTP	574	HTTP/1.1 200 OK (text/css)
1549	11.645608	2402:3a00:4632:ef14::64	ff9b::d7e:6584	HTTP	438	GET /assets/vendor/fancybox/css/jquery.fancybox.min.css HTTP/1.1
1553	11.645904	64:ff9b::d7e:6584	2402:3a00:4632:ef14::64	HTTP	144	HTTP/1.1 200 OK (text/css)
1557	11.647512	2402:3a00:4632:ef14::64	ff9b::d7e:6584	HTTP	458	GET /assets/vendor/owlcarousel/css/owl.carousel.min.css HTTP/1.1

Frame 1547: 574 bytes on wire (4592 bits), 574 bytes captured (4592 bits) on interface \Dev A

Ethernet II, Src: 78:30:22:85:0f:64 (78:30:22:85:0f:64), Dst: LiteonTechno_13:bc:bb (44:6d:00:10:00:00)

Internet Protocol Version 6, Src: 64:ff9b::d7e:6584, Dst: 2402:3a00:4632:ef14:25ad:6d0c:ee0

Transmission Control Protocol, Src Port: 80, Dst Port: 51553, Seq: 5381, Ack: 348, Len: 580

[5 Reassembled TCP Segments (580 bytes): #1541(1340), #1542(1340), #1544(1340), #1545(1340)]

Hypertext Transfer Protocol

- HTTP/1.1 200 OK\r\n
- Response Version: HTTP/1.1
- Status Code: 200
- [Status Code Description: OK]
- Response Phrase: OK
- Date: Sun, 16 Aug 2026 13:08:14 GMT\r\n
- Content-Type: text/css\r\n
- Content-Length: 5422\r\n
- Connection: keep-alive\r\n
- Last-Modified: Tue, 14 Jun 2022 13:30:25 GMT\r\n
- Accept-Ranges: bytes\r\n
- ETag: "abfada8f27fd81:0"\r\n
- Server: Microsoft-IIS/10.0\r\n
- Access-Control-Allow-Origin: *\r\n
- Access-Control-Allow-Headers: Content-Type, Accept, w-api-key\r\n

Frame (574 bytes) Reassembled TCP (580 bytes)

Packets: 9768 - Displayed: 150 (1.5%) - Dropped:

9. What is the status code and phrase returned from the server to your browser?(http.response.code)

➔ Status code:200

➔ Response phrase:OK

The image shows a Wireshark packet capture of an HTTP transaction. The top pane displays a list of packets, with packet 1288 selected. The middle pane shows the details of the selected packet, which is an HTTP 200 OK response. The bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
35	4.111086	2402:3a00:4632:ef14::64	ff9b::1739:e897	HTTP	287	GET /en-us/livetile/preinstall?region=IN&apple=CS&AS&B&A2009948508f971e10476512021FE368F&P=Thres
44	4.155252	64:ff9b::1739:e897	2402:3a00:4632:ef14::64	HTTP/1.1	1288	HTTP/1.1 200 OK
610	10.658393	2402:3a00:4632:ef14::64	ff9b::d7e:0504	HTTP	587	GET / HTTP/1.1
1417	11.580960	2402:3a00:4632:ef14::64	ff9b::d7e:0504	HTTP	413	GET /assets/css/responsive.css HTTP/1.1
1418	11.590212	2402:3a00:4632:ef14::64	ff9b::d7e:0504	HTTP	425	GET /assets/css/odometer-theme-default.css HTTP/1.1
1419	11.590504	2402:3a00:4632:ef14::64	ff9b::d7e:0504	HTTP	439	GET /assets/vendor/font-awesome/css/font-awesome.min.css HTTP/1.1
1420	11.590952	2402:3a00:4632:ef14::64	ff9b::d7e:0504	HTTP	457	GET /assets/vendor/themify-icons/css/themify-icons.css HTTP/1.1
1421	11.591184	2402:3a00:4632:ef14::64	ff9b::d7e:0504	HTTP	425	GET /assets/vendor/animate/animate.min.css HTTP/1.1
1547	11.645180	64:ff9b::d7e:0504	2402:3a00:4632:ef14::64	HTTP	574	HTTP/1.1 200 OK (text/css)
1548	11.645660	2402:3a00:4632:ef14::64	ff9b::d7e:0504	HTTP	438	GET /assets/vendor/fancybox/css/jquery.fancybox.min.css HTTP/1.1
1555	11.645984	64:ff9b::d7e:0504	2402:3a00:4632:ef14::64	HTTP	144	HTTP/1.1 200 OK (text/css)
1557	11.647512	2402:3a00:4632:ef14::64	ff9b::d7e:0504	HTTP	458	GET /assets/vendor/owlcarousel/css/owl.carousel.min.css HTTP/1.1

Packet Details:

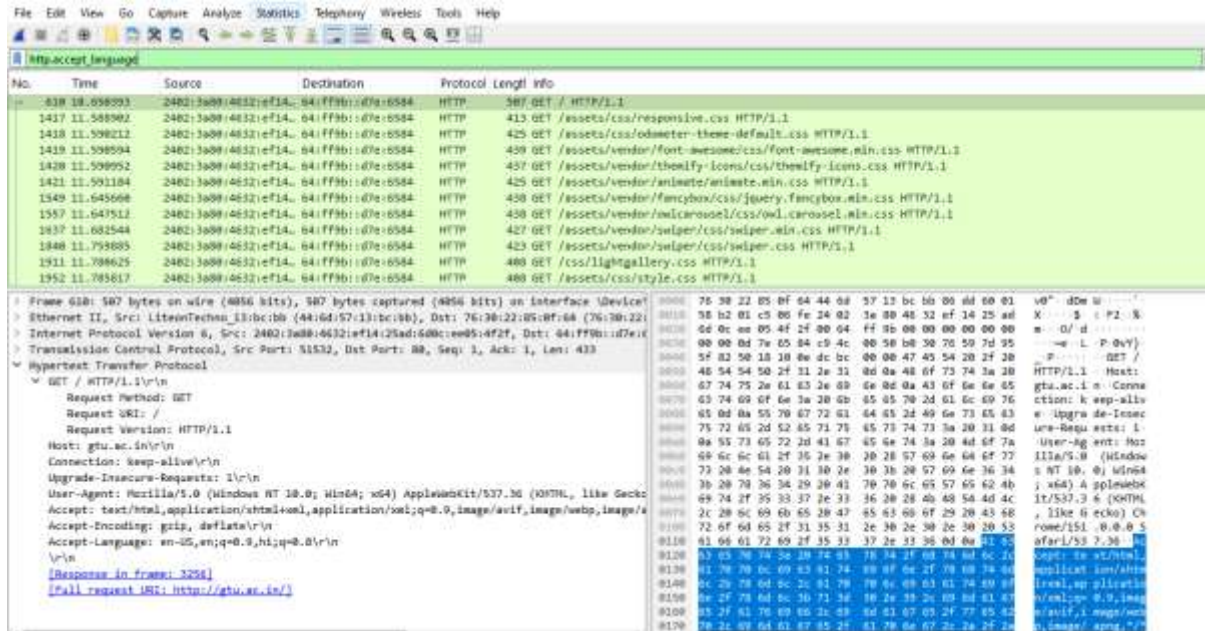
Frame 44: 1288 bytes on wire (9604 bits), 1288 bytes captured (9604 bits) on interface \Dev...
Ethernet II, Src: 70:30:21:09:0f:d4 (70:30:21:09:0f:d4), Dst: LiteonTechno 13:bc:bb (44:6d:13:bc:bb)
Internet Protocol Version 6, Src: 64:ff9b::1739:e897, Dst: 2402:3a00:4632:ef14:25ad:609c:ee
Transmission Control Protocol, Src Port: 80, Dst Port: 81600, Seq: 6781, Ack: 214, Len: 113
[6 Reassembled TCP Segments (7834 bytes): #37(1340), #38(1540), #40(1340), #41(1340), #43(1340)]
Hypertext Transfer Protocol
 HTTP/1.1 200 OK\r\n
 Response Version: HTTP/1.1
 Status Code: 200
 [Status Code Description: OK]
 Response Phrase: OK
 Content-Type: application/xml; charset=utf-8\r\n Access-Control-Allow-Credentials: true\r\n Access-Control-Allow-Headers: TicketType,RequestContinuationKey,AuthToken,Content-Type
 Access-Control-Allow-Methods: PUT,PATCH,POST,GET,OPTIONS,DELETE\r\n Access-Control-Allow-Origin: *.com\r\n [-] Access-Control-Expose-Headers: TicketType,RequestContinuationKey,AuthToken,Content-Type
 OOO-AuthenticatedWithJwtFlow: False\r\n OOO-UserType: Anonymous\r\n XAP-Workflow-Version: 2.436.0\r\n OOO-DebugId: 100x5002-3584-45fc-a7f5-f652f1180da2|2026-08-16T13:06:47.17325032|fabriz_mai

Raw Data:

0000 44 68 57 13 bc bb 76 38 22 85 0f 64 06 d8 69 80 0000 04 82 80 38 00 04 ff 9b 00 00 00 00 00 00 00 0000 17 39 e8 97 24 02 3a 00 46 32 ef 14 25 ad 0000 6d bc ee 85 4f 2f 00 50 f0 ad 01 f6 03 e1 23 0e0000 5f be 58 18 01 f5 bb 20 00 00 00 00 00 00 0000 13 38 76 35 30 2f 33 2a 78 66 87 3f 61 22 2f 3e0000 4d 8e 28 28 20 20 20 20 20 20 20 20 20 20 2000 74 28 68 68 6e 74 28 61 6c 68 67 6e 74 22 63 650000 6e 74 65 72 22 3e 35 31 c2 60 3c 2f 74 65 78 740000 4e 6d 68 28 28 20 20 20 28 28 28 28 28 3c 74 280000 78 74 28 68 68 6e 74 28 75 74 70 6c 62 38 22 630000 61 78 74 68 6f 6e 73 70 62 74 6c 63 21 20 68 680000 6e 74 28 61 6c 68 67 6e 3d 22 63 65 6e 74 65 720000 22 3e 3f 39 c2 60 3c 2f 74 65 78 74 3e 68 6e 380000 28 28 28 28 28 28 28 28 2f 73 75 62 67 72 6f 750000 70 3e 6d 6e 28 28 20 20 28 28 20 20 3c 73 75 620000 67 72 67 75 70 28 68 6e 6e 74 28 73 65 68 67 680000 74 3d 21 31 30 22 3e 6e 6e 28 28 28 28 28 280000 28 28 28 3c 74 65 78 74 3d 63 65 6e 74 2d 61 6e0000 67 67 6e 3d 21 61 68 6e 74 65 72 22 3e 54 75 650000 6e 2f 73 65 70 74 3e 6e 6e 28 28 28 28 28 280000 28 28 28 3c 6d 6d 62 67 65 28 68 6e 74 2d 610000 6c 67 67 6e 3d 21 61 68 6e 74 65 72 22 3e 73 72

10. What languages does your browser indicate to the server that it can accept? Which header line is used to indicate this information?

http.request or http.accept_language

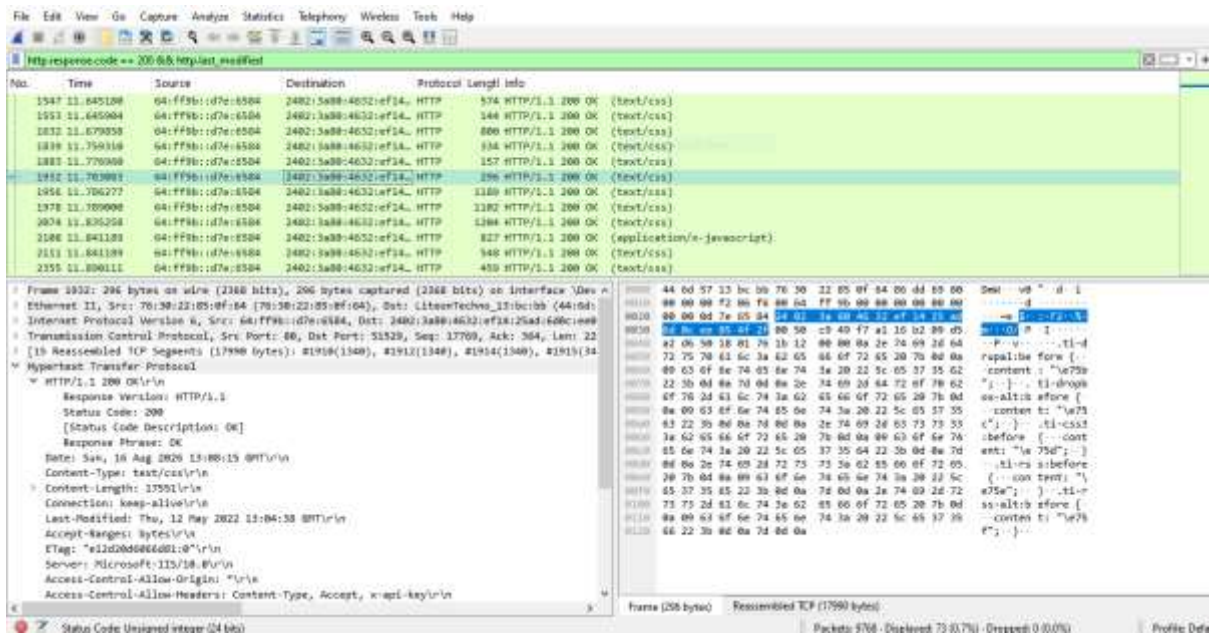


ANS: Accept language:en-US,en;q=0.9,hi;q=0.8\r\n
Accept language is header line

11. When was the HTML-file, that you have retrieved, last modified at the server? Which header line is used to indicate this information?

(When was this webpage last updated on the server)

http.response.code == 200 && http.last_modified



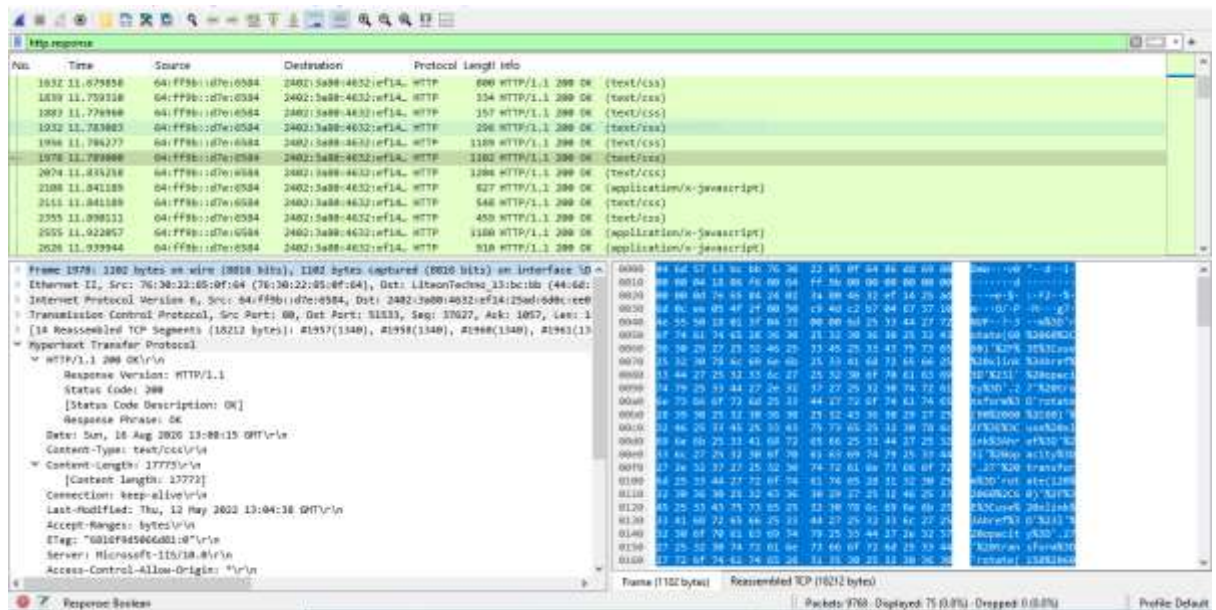
ANS: Last-Modified:Thu,12 May 2022 13:04:38

12. How many bytes of content (size of file) are returned to your browser? Which header line is used to indicate this information?

(When the server sends the webpage (HTML file) to your browser, what is the size of that file?)

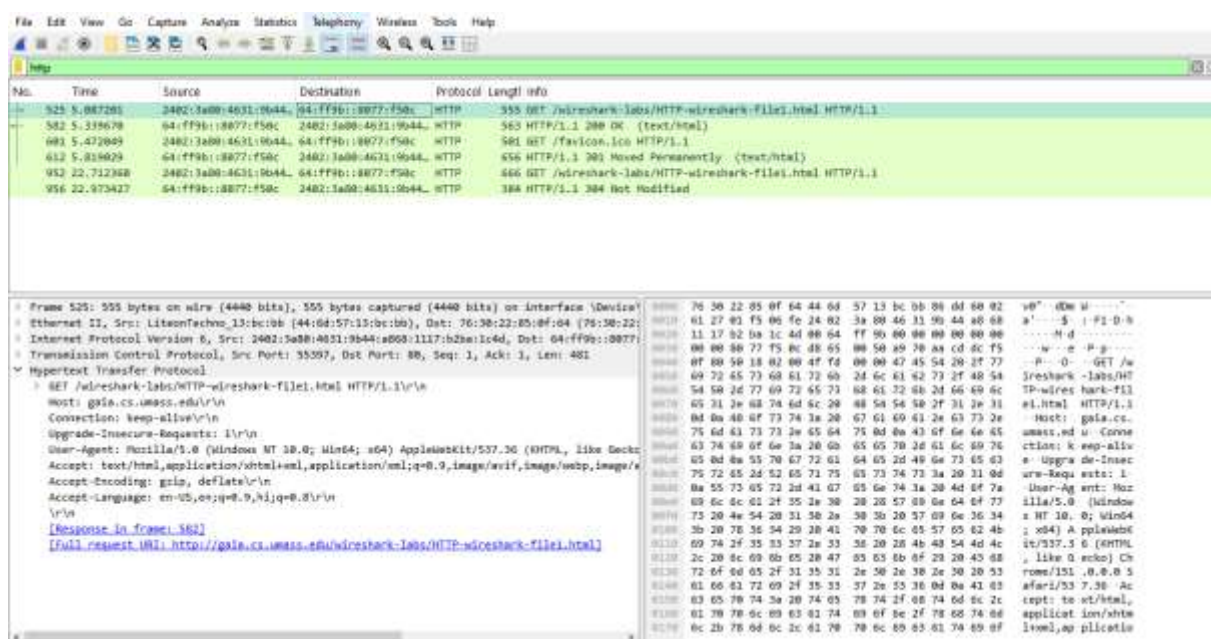
ANS: Content-Length: 17551\r\n

Header line : Content length



HTTP Conditional GET/Response Interaction

13. Start up the Wireshark packet sniffer, and make sure that “http” is in the display filter, so that only captured HTTP messages will be displayed in the packet-list pane.
14. Enter the following URL into your browser:
<http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html>
Your browser should display a very simple HTML file.
15. Quickly enter the same URL into your browser again (or simply select the refresh button on your browser).
16. Stop Wireshark packet capture.



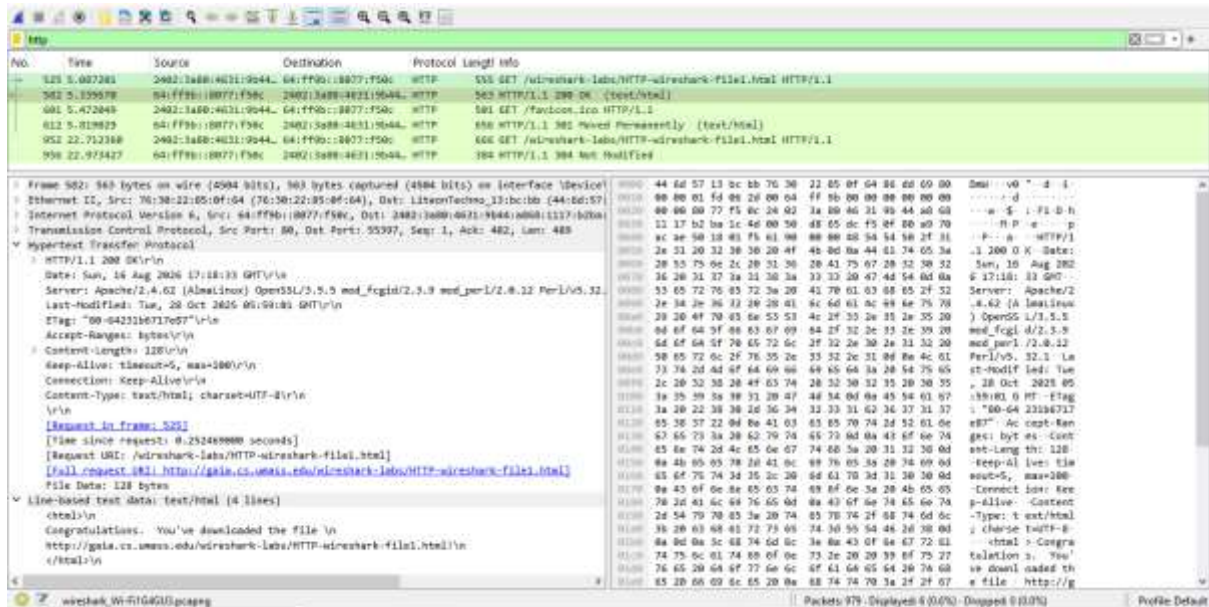
Answer the following questions. Please explain how you can find the answers

Ans: No. If-Modified-Since is not present because the browser is requesting the page for the first time. It has no previous copy of the page, so it simply asks the server to send the page.

18. Inspect the contents of the server response. Has the server explicitly returned the contents of the file? How can you tell?

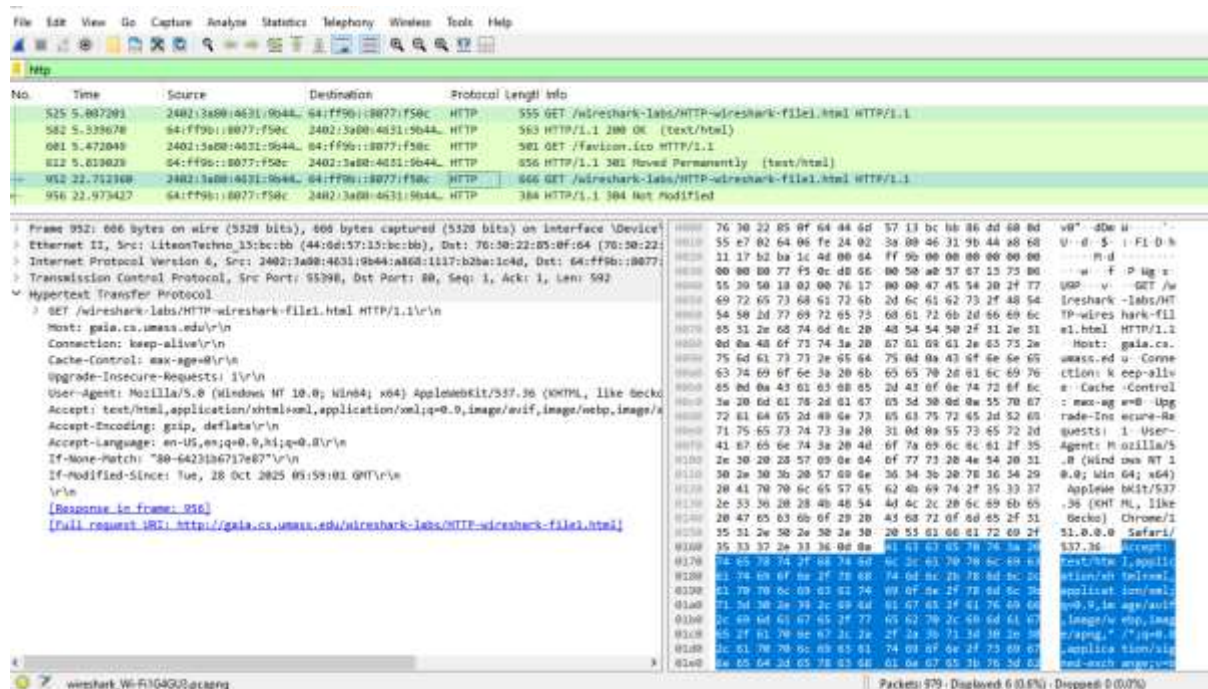
Ans :- Yes, the server has returned the contents of the file.

Extend line based text data: Congratulation you've download the file



19. Now inspect the contents of the second HTTP GET request from your browser to the server. Is there an “IF-MODIFIED-SINCE:” header line in the HTTP GET message? If so, what information follows the “IF-MODIFIED-SINCE:” header line? (second link - yes)

Ans: Yes. If-Modified-Since is present because the browser already has a copy of the page. It asks the server whether the page has changed since the last time it was downloaded.



(ans of next que) If it has not changed, the server replies with **304 Not Modified**.

If **yes**, the server sends the updated file (**200 OK**).

20. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Has the server explicitly returned the contents of the file? Explain.

- ➔ Status code: 304
- ➔ Response phrase: Not modified

No, the server has not returned the contents of the file as it is the response of the second request sent by the browser.

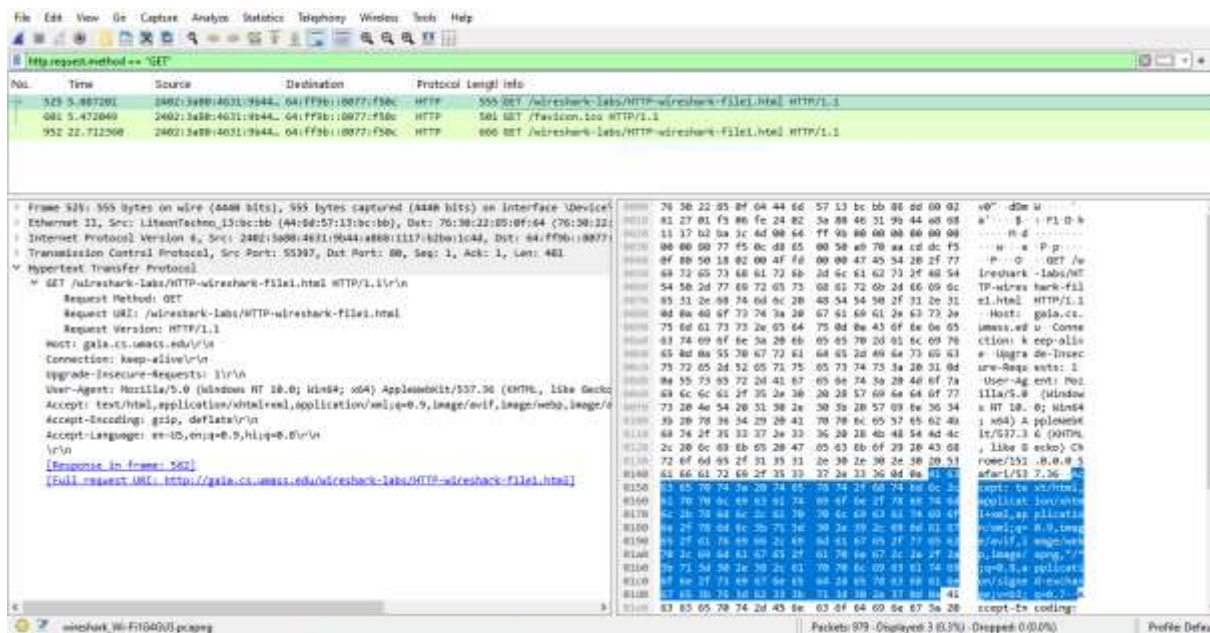
The image shows a Wireshark packet capture of an HTTP transaction. The packet list on the left shows five packets. The selected packet is packet 5, an HTTP GET request for /wirehark-labs/HTTP-wireshark-file1.html. The packet details pane on the right shows the response structure for the selected packet, which is an HTTP 304 Not Modified response. The status code is 304 and the status phrase is 'Not Modified'. The response is an empty body, indicating that the server has not returned the contents of the file. The packet bytes pane on the right shows the raw data of the response, which is an empty body.

Frame 5: 384 bytes on wire (3072 bits), 384 bytes captured (3072 bits) on interface libvirt0
Ethernet II, Src: 70:39:22:05:0f:84 (70:39:22:05:0f:84), Dst: LinuxTechon 13:bc:b6 (44:6d:57:13:bc:b6)
Internet Protocol Version 6, Src: 64:ff9b:1007:f50c, Dst: 2002::3a8b:6631:9b44:a060:1117:82ba
Transmission Control Protocol, Src Port: 80, Dst Port: 55396, Seq: 1, Ack: 593, Len: 518
Hypertext Transfer Protocol
HTTP/1.1 304 Not Modified
Response Version: HTTP/1.1
Status Code: 304
[Status Code Description: Not Modified]
Response Phrase: Not Modified
Date: Sun, 16 Aug 2026 17:18:50 GMT
Server: Apache/2.4.62 (Ubuntu) OpenSSL/3.0.3 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32
last-Modified: Tue, 20 Oct 2025 05:59:01 GMT
ETag: "00-94231b6717e57"
Accept-Ranges: bytes
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
[Request to Frame: 957]
[Time since request: 0.283867088 seconds]
[Request URI: /wirehark-labs/HTTP-wireshark-file1.html]
[Full request (87): http://64.ff9b:1007:f50c:2002::3a8b:6631:9b44:a060:1117:82ba/HTTP-wireshark-file1.html]

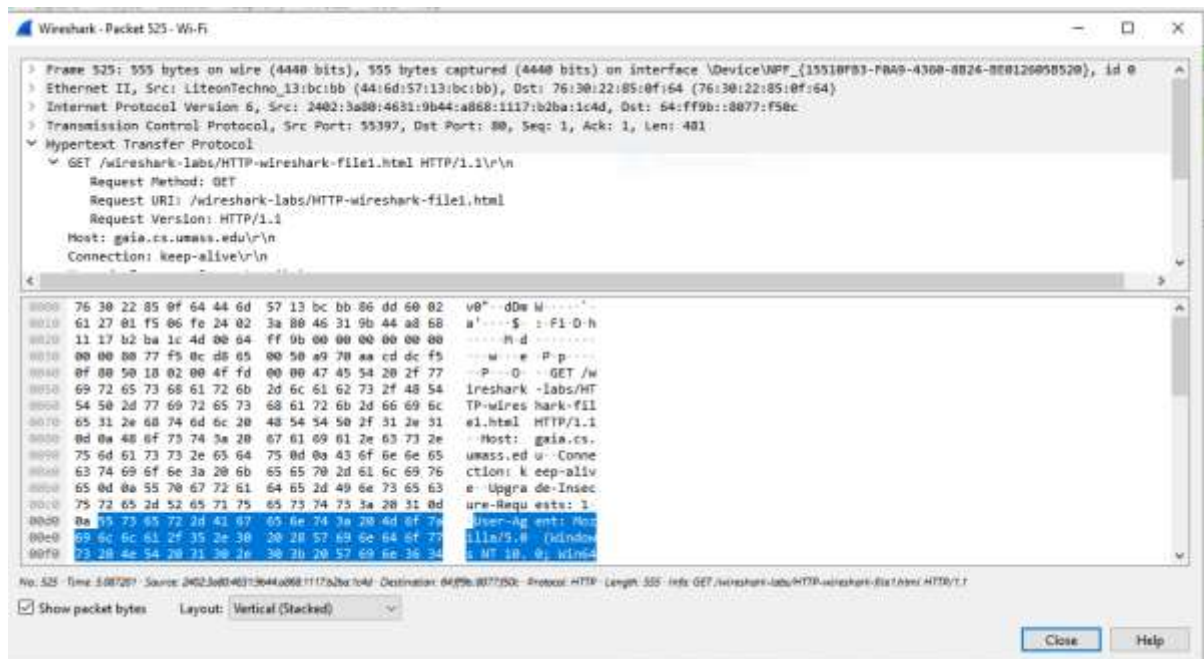
21. How many HTTP GET request messages has your browser sent? Which packet in the trace contains the request for the Bill of Rights?
 http.request.method=="GET"

(The **packet number** of that HTTP GET packet is the answer.)

Total HTTP GET request messages :- 3



->Packet num is 525 of get request

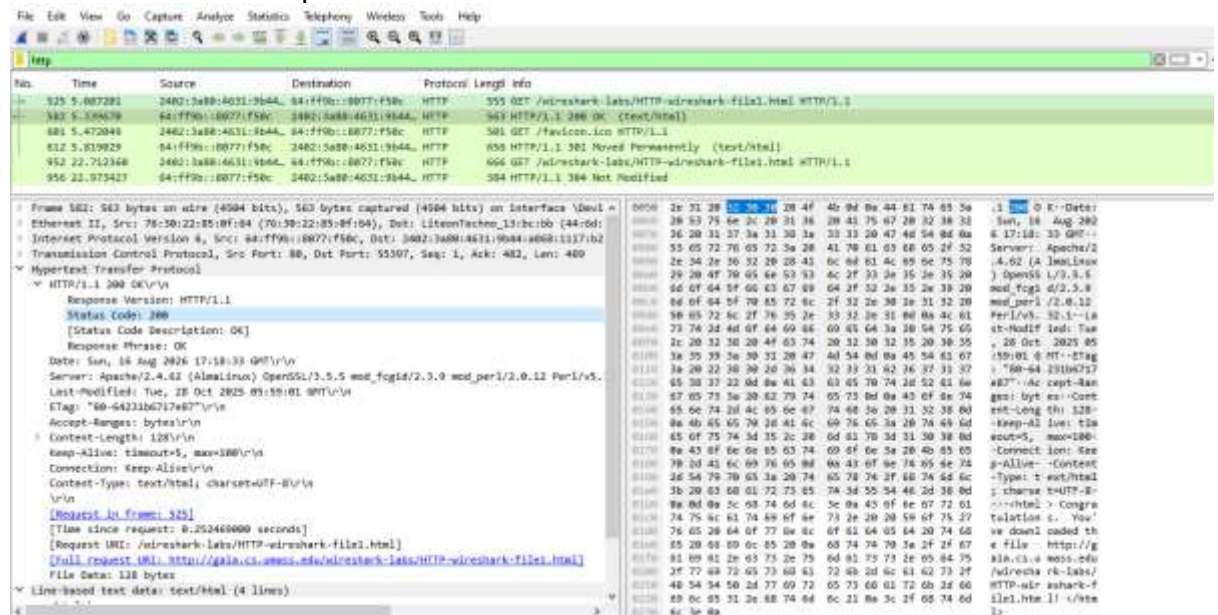


22. Which packet in the trace contains the status code and phrase associated with the response to the HTTP GET request? What is the status code and phrase in the response?

Or

(Which packet contains the server's response to the HTTP GET request? What is the status code and status phrase in that response?)

Ans: Status Code :- 200 , Response Phrase :- OK
Packet no of server response is : 582



23. Discuss http GET and RESPONSE.

HTTP GET Request:

An **HTTP GET** request is sent by the **browser (client)** to the **server** to request a web page or other resource. It contains information such as the requested URL, HTTP version, and request headers.

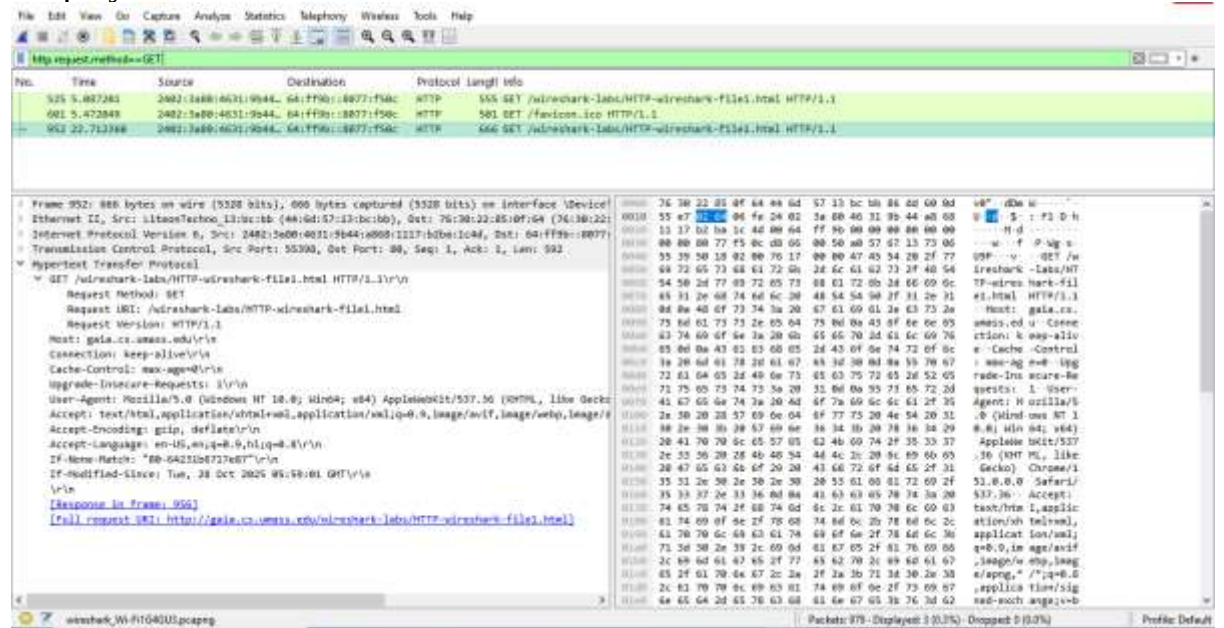
HTTP Response:

An **HTTP Response** is sent by the **server** back to the **browser** after receiving the GET request. It contains the **status code** (such as **200 OK** or **304 Not Modified**), response headers, and, if required, the requested web page or file.

24.Count the total number of HTTP GET requests.

Ans.: Enter in filter: http.request.method==GET , then count total ----- marked.

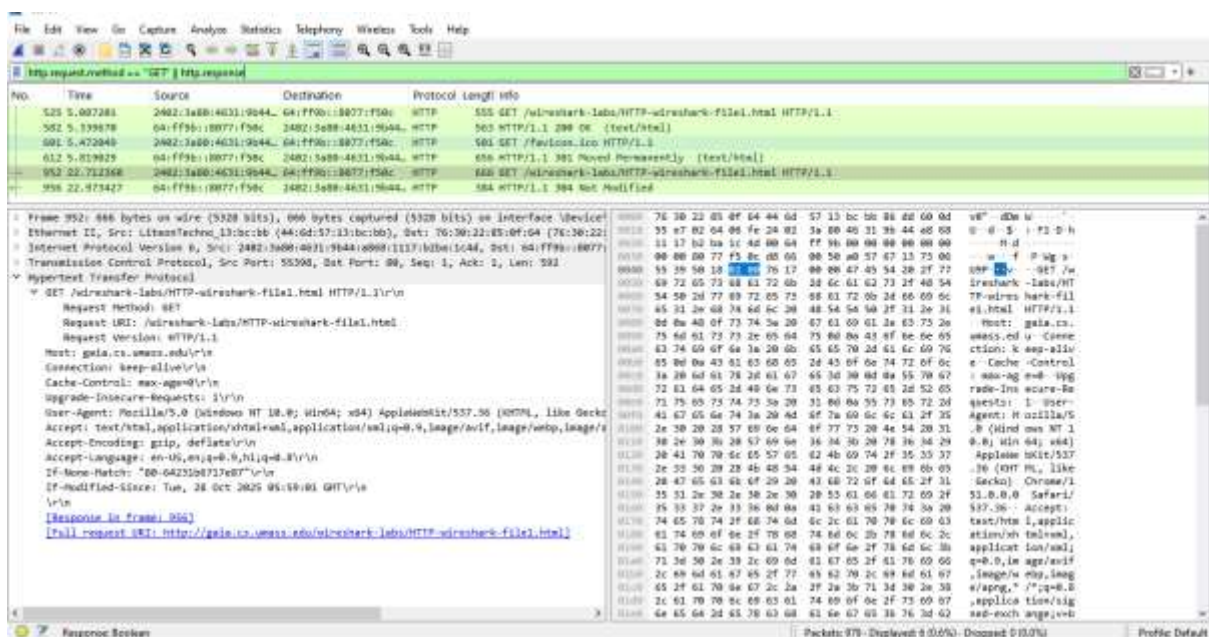
Displayed: 3



25. How much time elapsed between the first HTTP GET request from client and the HTTP response (OK) from server?

**Ans.: $http.request.method == GET \parallel http.response$
(Response Time – GET Time = Elapsed Time)**

(How much time did the server take to reply after the browser sent the first HTTP GET request?)



Response time is: 5.339670

Request time is: 5.087201

Elapsed time is: 0.312469

26. Identify the source and destination ports used to transfer the first HTTP GET request

No.	Time	Source	Destination	Protocol	Length	Info
525	0.000000	192.168.1.100	192.168.1.1	HTTP	555	GET /index.php?id=1 HTTP/1.1
526	0.000000	192.168.1.1	192.168.1.100	HTTP	563	HTTP/1.1 200 OK (text/html)
527	0.000000	192.168.1.100	192.168.1.1	HTTP	561	401 /favicon.ico HTTP/1.1
528	0.000000	192.168.1.1	192.168.1.100	HTTP	656	HTTP/1.1 302 Moved Permanently (text/html)
529	0.000000	192.168.1.100	192.168.1.1	HTTP	666	401 /index.php?id=1 HTTP/1.1
530	0.000000	192.168.1.1	192.168.1.100	HTTP	304	HTTP/1.1 304 Not Modified

Frame 525: 555 bytes on wire (4440 bits), 555 bytes captured (4440 bits) on interface 0
Ethernet II, Src: MikrotikTech (88:63:83:12:12:12), Dst: 192.168.1.1 (08:00:27:00:00:00)
Internet Protocol Version 4, Src: 192.168.1.100, Dst: 192.168.1.1
Transmission Control Protocol, Src Port: 55397, Dst Port: 80, Seq: 1, Ack: 1, Len: 481
- Source Port: 55397 - Destination Port: 80 [Stream Index: 5] [Stream Packet Number: 4] [Conversation completeness: (complete, with data [3])] [TCP Segment Len: 481] - Sequence Number: 1 (relative sequence number) - Sequence Number (raw): 284273201 [Next Sequence Number: 482 (relative sequence number)] - Acknowledgment Number: 1 (relative ack number) - Acknowledgment Number (raw): 370706788 - Offset: 0 (header length: 20 bytes (3)) - Flags: RST (RST, ACK) - Window: 512 [Calculated window size: 131072] [Window size scaling factor: 256] - Checksum: 0x0000 [unverified] [Checksum Status: Unverified] - Urgent Pointer: 0 [Timestamps] [SEQ/ACK analysis] - TCP payload (481 bytes)

➔ Extend transmission protocol:
Source port:55397
Destination port:80

27. Identify the client and server ip address in the first HTTP GET request?

Ans.: Apply filter `http.request.method==GET`, see the IP, within this you will find source ip address and destination ip address.

Extend internet protocol

The image shows a Wireshark packet capture of an HTTP GET request. The packet list at the top shows a GET request for `/favicon.ico` from source IP `192.168.1.100` to destination IP `192.168.1.1`. The packet details pane shows the HTTP request structure, including the request line `GET /favicon.ico HTTP/1.1`, the host `192.168.1.1`, and the user-agent `Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.114 Safari/537.36`. The packet bytes pane shows the raw data of the request, including the request line, host, and user-agent.

28. Identify the client and server MAC address in the first HTTP GET request?

Ans.: Apply filter `http.request.method==GET`, see the Ethernet, within this you will find source MAC and destination MAC.

Extend Ethernet:

